Halo communication in the mEVP sea-ice solver of the FESOM2 GPU port

N. Koldunov, 6 August 2026 — summary of measurements

Three ways of exchanging the halo less often

mEVP advances the ice momentum in 120 sub-cycles per model time step, and the ice-velocity halo is at present exchanged after each of them. We implemented and measured three alternatives.

Delayed exchange — exchange every K-th sub-cycle and accept a halo up to $K-1$ sub-cycles old. This is the largest saving of the three, but it leaves the sub-domain boundaries visible in the ice field. *Wide halo* (Danilov) — exchange every K-th sub-cycle, and recompute a K-deep ring of ghost cells in between so that no value is stale; the result reproduces the every-sub-cycle trajectory bit for bit. *Divergence form* (Danilov) — carry $\nabla\cdot\sigma$ at nodes rather than σ at elements; it agrees with $\nabla\cdot\sigma$ to 10^{-15} relative, and its effect on run time differs between the two backends.

Cost, measured against the model step and against the ice

Each number is given twice. Sea ice is 5 to 13% of a CPU model step, so an ice-only change is divided by about ten before it reaches the step, and the share of the ice cost (ice, ice dynamics and ice advection, computation together with time spent waiting for communication) is the measure appropriate to a sea-ice scheme.

		GPU		CPU	
		ice cost	model step	ice cost	model step
delayed exchange	K=2	−32 to −39%	−8.3 to −9.3%	−2 to −18%	−0.1 to −2.9%
delayed exchange	K=4	−37 to −58%	−8.5 to −14%	−5 to −25%	−0.3 to −3.6%
wide halo, standard form	K=8	−11 to −24%	−2.2 to −5.7%	never profitable	
wide halo, divergence form	K=8	+5 to −29%	+0.2 to −6.8%	never profitable	
wide halo, either form	K=2	−15% †	+1.7 to −3.8%	never profitable	
divergence form alone	—	+2 to +17%	0 to +1.6%	−2 to −8.5%	−0.5 to 0%

GPU ranges span the meshes at the node count where each still gains from more GPUs: CORE2 on 1 node (4 GPUs), fArc on 4, DARS on 8, NG5 on 16 — the wide-halo rows cover the first three, NG5 being still in the queue. CPU ranges span 128 to 2048 cores on CORE2 and fArc. † one point only (DARS, 8 nodes); at K=2 the wide halo is a cost on fArc at 4 nodes and a saving at 16, so it needs the larger K.

On CPU the saving grows with core count (−5% of the ice at 128 cores, −25% at 2048); on GPU it falls as each GPU holds more of the mesh (−58% at 32 000 surface nodes per GPU, −37% at 460 000). The divergence form removes 2 to 8.5% of the ice cost from 512 cores upward, and is invisible in the model step only because of the denominator. On GPU it costs, because the sub-cycle there is limited by how often work is dispatched to the device rather than by arithmetic or memory traffic, so removing element arrays removes work that was not being paid for. Danilov's ∼20% for the divergence form with its double halo corresponds to our −19.1% of the ice cost for the same combination (fArc, 16 GPU nodes).

Why the delayed exchange is not usable

Under the delayed exchange the stale values lie on the one-cell boundary of each sub-domain, and the decomposition is fixed for the run, so the same nodes are perturbed at every step. On fArc with 16 GPUs over 60 days, $|\Delta$ ice thickness$|$ averaged on those boundaries is 2.5 times its interior value at K=4 and 1.7 times at K=2, against 0.85 for two identical runs and 0.87 for the exact wide halo. The difference is concentrated there from the first day and appears on maps as a network following the sub-domain edges. A polygonal structure in the ice field, in the shape of the domain decomposition, is not acceptable in a sea-ice model, and that ends the matter — the scheme's accuracy in the integral diagnostics is beside the point. Section 6.6 and Figure 4 of the full note give the profiles, the map and the controls.

Recommendation

For production we would use the exact wide halo at K=8, in the divergence form on the larger meshes and the classic form on the smaller ones. Either is bit-for-bit identical to the standard scheme, so neither requires an accuracy argument nor the check above; together they remove 11 to 29% of the ice cost on GPU. On CPU the wide halo is not profitable on any mesh we tried.

We do not regard the delayed exchange as usable, although it saves about three times as much: it places the domain decomposition in the ice field.

The divergence form should be judged in two roles. On its own it costs 2 to 8% of the ice on GPU and saves 2 to 8.5% on CPU at large core counts. As the carrier of the wide halo it is the better of the two forms wherever each GPU holds enough of the mesh: against the classic form it loses by 17 percentage points on CORE2 at 32 000 surface nodes per GPU, loses by 4 on fArc at 40 000, and **gains 5 on DARS at 99 000** — the mesh on which the wide halo pays most, and the configuration closest to the ones we would run. Everything sub-cycled is then a node quantity, and the halved message is what buys this. NG5, at 116 000 nodes per GPU, is queued and will test whether the trend continues.